Slide 1

Imprinting

When will a deletion cause different effects if it is inherited through father versus mother?

Turningpoint id: Larsen

Slide 2

Learning Objectives

- Explain the changes to imprinting status during the lifetime of a person
  - Explain that DNA sequences do not change
  - Explain the involvement of methylation and histone modifications in silencing of an area
- Explain the differences between male and female gametogenesis relative to imprinting
- Explain what uniparental disomy is
- Explain the involvement of imprinted genes in hydatidiform moles
- Explain the interaction of imprinted genes and deletions as well as uniparental disomy in Prader-Willi and Angelman syndromes
- Explain the involvement of imprinting in Beckwith-Wiedeman and Silver-Russel syndromes
- Compare and contrast the four syndromes
- Make appropriate predictions based on this knowledge in for example pedigrees.
- Reading: Emery's elements of medical genetics (15th ed) pp76-80
- Video on Angelman and Prader Willi syndromes: https://usmle-rx.scholarrx.com/video-player;video=2349

Slide 3

For Office Hours

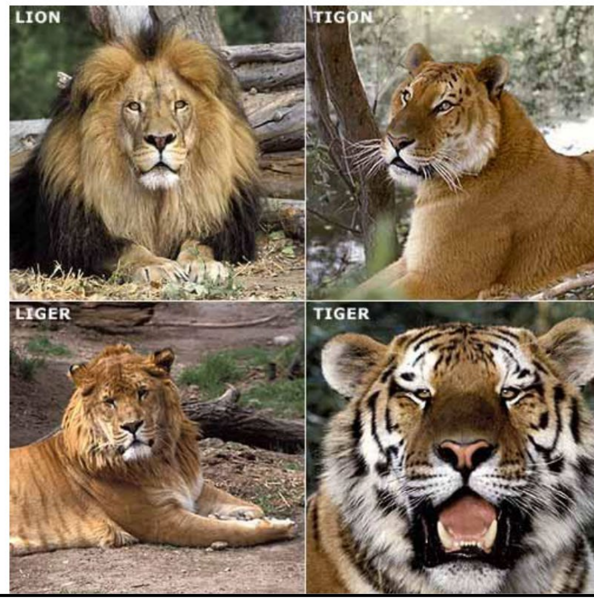
- Go to Canvas, settings and set time zone to AST (Georgetown is a good selection). Setting your device time zone is not enough.
- Then check the Canvas calendar for office hours.

Slide 4

The appearance of the offspring depends on who the father and mother were.

Lion X Lion

Tiger X Lion



Lion X Tiger

Tiger X Tiger

Slide 5

Syndromes for Today

Prader-Willi Syndrome

Beckwith Wiedeman Syndrome

Angelman Syndrome

Silver-Russell Syndrome

All images found using BING and the option of “Free to share and use”. BING never took me to the original page, so do not know where they came from.

Slide 6

Imprinting

- **Imprinting is normal**, present in many chromosomal regions in humans (~25)
- >100 genes listed as imprinted
  (http://www.geneimprint.com/site/genes-by-species)
- Imprinting is a modification to DNA/chromosomal proteins that is erasable.
  = Parent-of-origin-specific silencing of a gene
- **Imprinted = ONE allele inactive**!

Imprinting is one example of epigenetics

You already heard about epigenetics in X-chromosome inactivation, mechanism is quite similar.

These two statements say the same:
-Maternally imprinted gene
-Imprinted gene, paternally expressed.
In both of these, one allele is the imprinted, inactive allele (the one from mother) and the other is expressed, **not** inactive, the one from father.

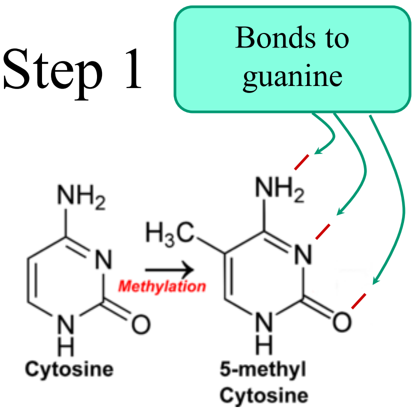
Inheritance of Imprinting

- Imprinting is **NOT** inherited in the normal sense of the word!
- During gametogenesis:
  - imprinting is erased.
  - new imprinting is added according to sex of parent

Slide 8

Mechanism Step 1

- **Methylation** of CG base-pairs
- Reversible
- Does not interfere with base pairing
- Symmetrical
- During replication, one strand is template for methylation of other strand



5′-AGCGGTGG-3′
3′-TCGCCACC-5′

Methyl group added to location 5 in Cytosine. As can be seen, this is not interfering with binding to Guanine so base pairing is normal. The location of the methyl group is accessible to proteins binding to the DNA double helix.

In the DNA sequence, the Cs shown in red could be methylated. Notice how symmetrical the pattern is.

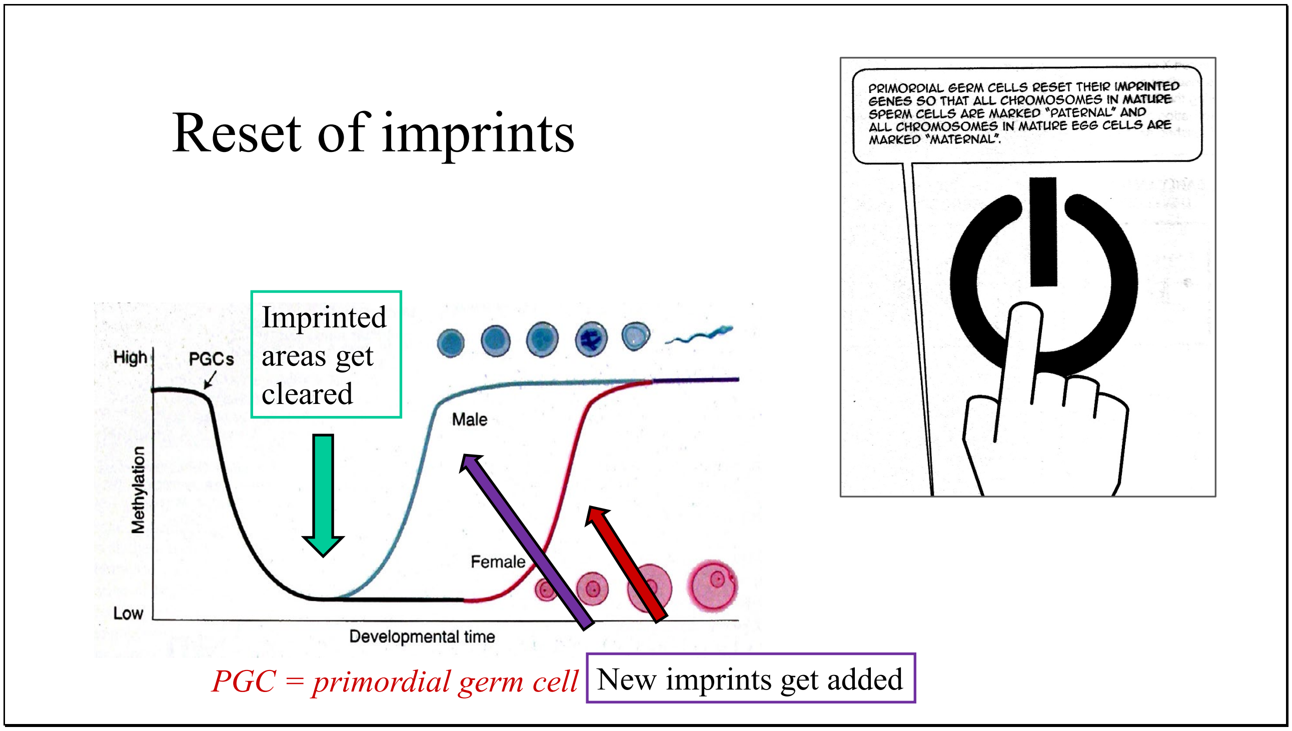
Slide 9

Mechanism: beyond methylation

- Modified histones specific for silenced genes localize
- At least two non-histone proteins help maintain status

Essentially similar to the changes that happen after methylation in areas silenced by X-chromosome inactivation, even though the initiating factors (leading to methylation) are different.

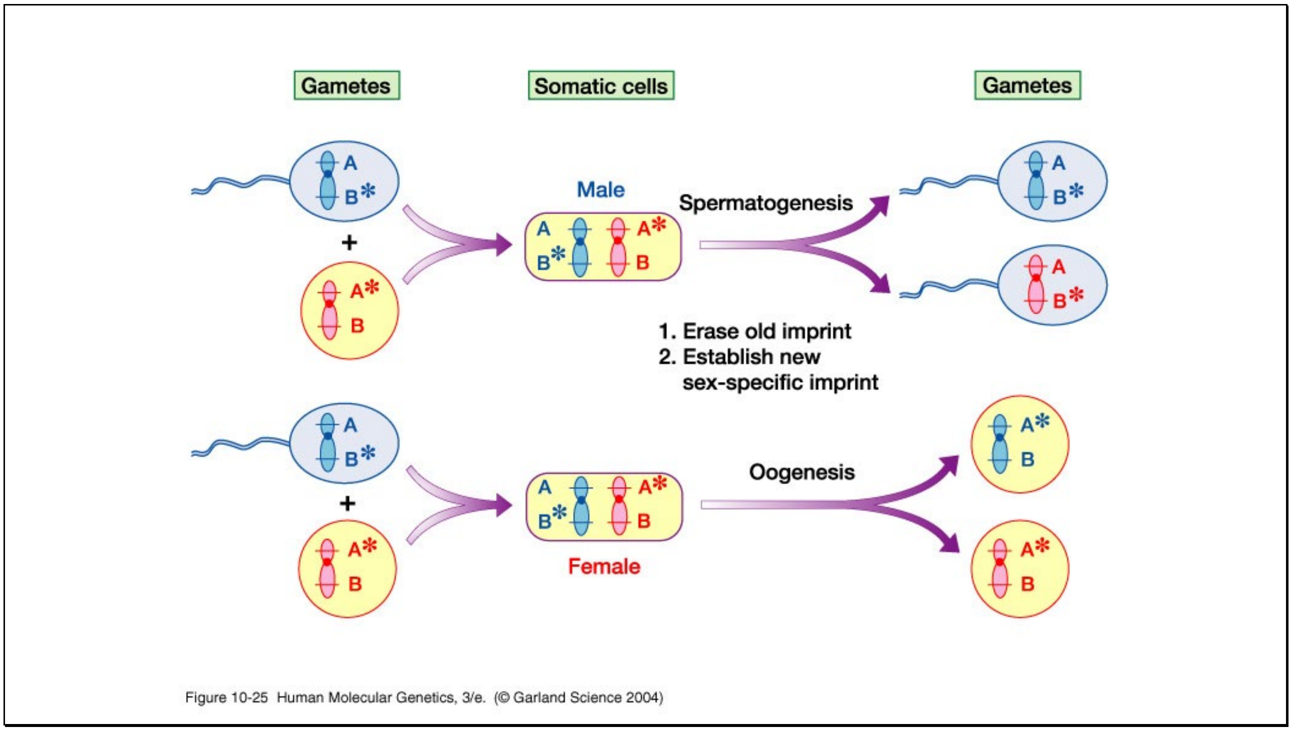
Slide 10



During gametogenesis the imprinted areas are cleared of all signs of imprinting including the methylation. Shortly thereafter, brand new imprints are added to all chromosomes in the gamete precursors.
Errors in the processes described here can sometimes lead to disease state in the child.
One way that this process can go wrong is if an imprint is not removed in the early stage.

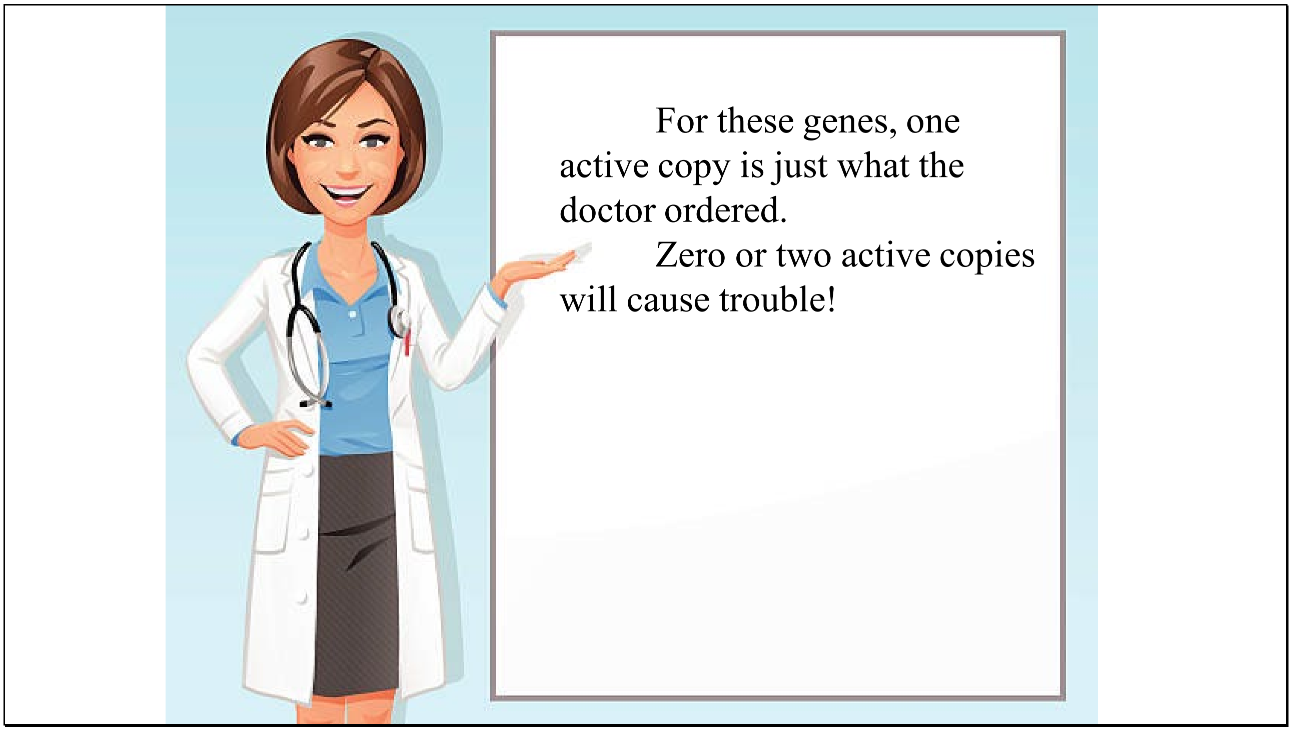
Figure at the right from Ennis and Pugh 2017. Introducing Epigenetics - a graphic guide. Icon books ldt.
This book is an easy read about some of the newer genetics topics.

Slide 11



All sperm identical (world wide)
All eggs identical (world wide)
Eggs vs sperm: different from each other
Therefore, it is clear that imprinting is removed during gametogenesis before being applied again to all chromosomes (as demonstrated on other slides). Normal development only happens if the zygote contains one chromosome imprinted one way and the other chromosome imprinted with the other pattern (within the chromosome pair shown here).

Slide 12



The background clipart was found using search for items “free to use and share”

Prader Willi and Angelman syndromes

Prader-Willi Syndrome

- Presents with hypotonia in infancy (even prenatal)
  - Includes difficulty feeding
- Developmental delay from day 1
- Intellectually: mild disability
  - On top of that: learning disability, poor academic performance
  - Some cases described as “not having a language”
- Hypogonadism, frequent cryptorchidism in males, underdeveloped clitoris in females
- Hypopigmentation in 30-50% of patient

UpToDate and GeneReviews (at NIH) main sources for symptoms

An introduction to the Prader-Willi syndrome (and a little on Angelman) is found at https://www.youtube.com/watch?v=nYvm4Rsh7t8

Slide 15

Prader-Willi Syndrome

- Growth hormome deficiency is frequent
- Appetite increases in mid childhood often with inability to feel full
  - Results in morbid obesity and type 2 diabetes unless parental control
- Behavior: temper tantrums, stubbornness, controlling and manipulative behavior, compulsivity, and difficulty with change
- Incidence 1/10,000.
- Loss of expression of several genes in 15q11.2-13

Image from http://zygote.swarthmore.edu/chrom3a.html
site no longer functional

UpToDate and GeneReviews (at NIH) main sources for symptoms
Genes involved: SNRPN = small nuclear ribonucleoprotein polypeptide N, predominantly expressed in brain
NDN = Necdin

Slide 16

Angelman Syndrome

- Developmental delay noticable from ½ year of age
- Progressive, several years to full syndrome
- Gait ataxia, tremoulous hands, intellectual disability, speech impairment, inappropritate laughing/smiling/exitability
  - Microcephaly, seizures frequently seen
- Incidence 1/20,000.
- Loss of expression of UBE3A in 15q11.2

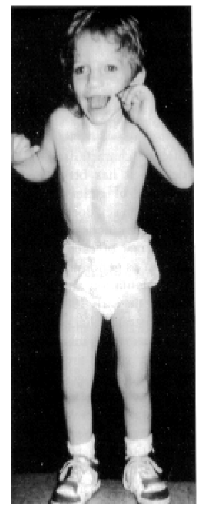


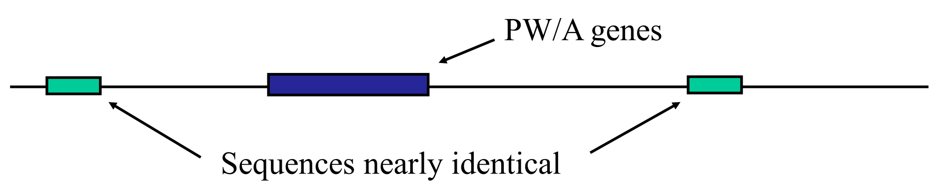
Image from http://zygote.swarthmore.edu/chrom3a.html (no longer functional)

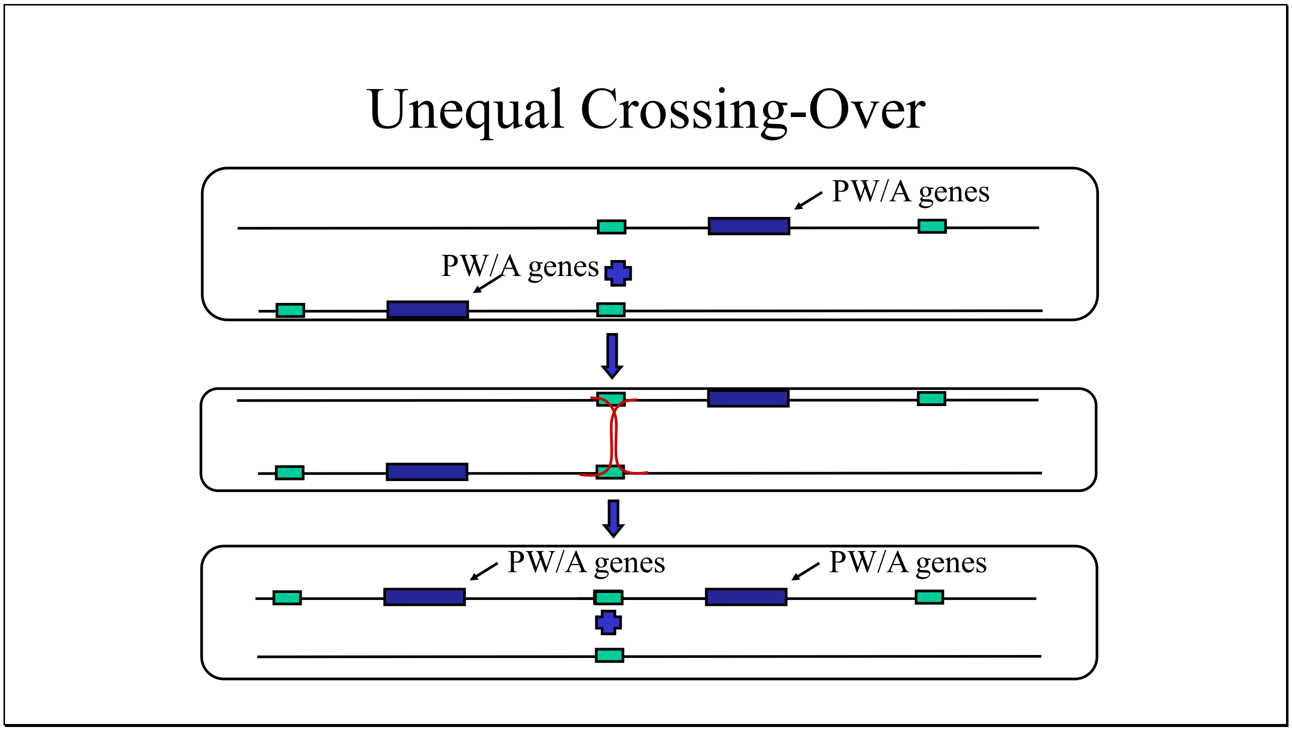
UBE3A = ubiquitin-protein ligase E3A

Slide 17

Causes of Prader-Willi or Angelman

- Most often a *de novo* deletion (70-80 % of cases).
- The most common deletions are 3-4 mb in size.
- Similar sequences → unequal crossing over.





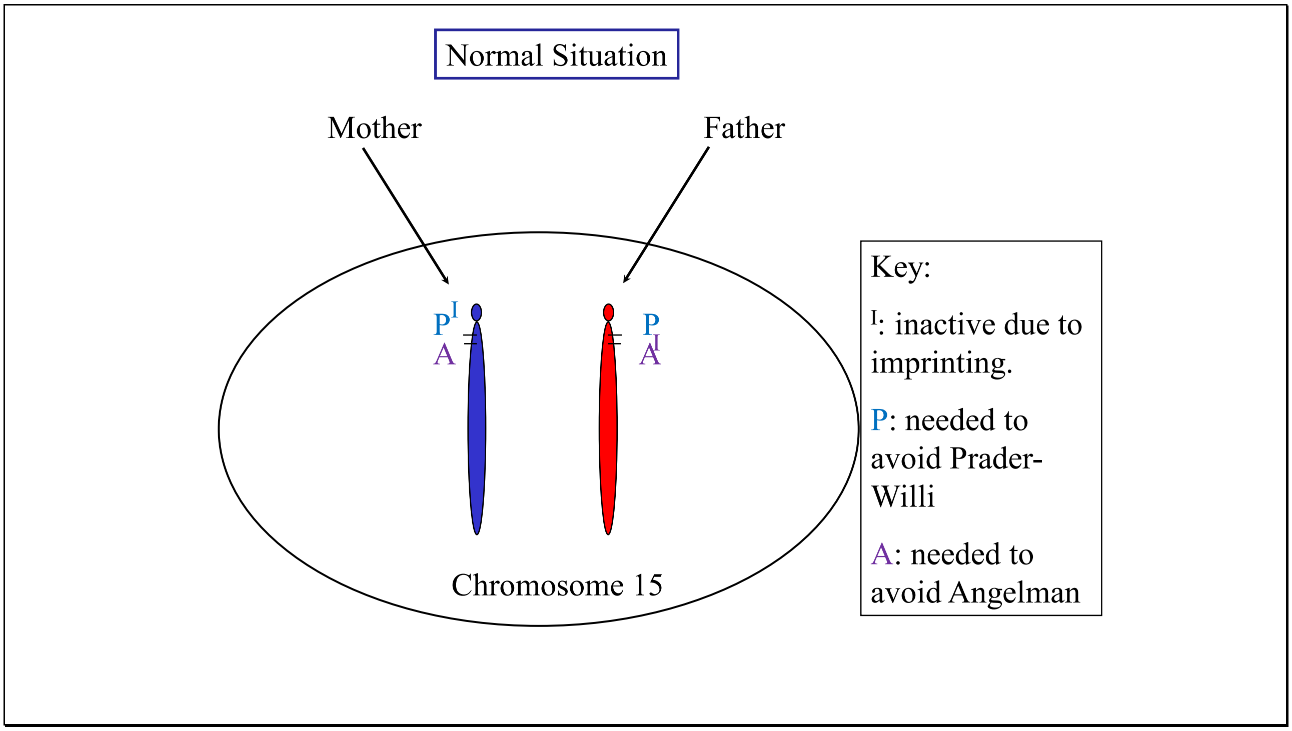
In the top image is shown an alignment of a chromosome pair which is not correct, these are not in register. This alignment is driven by the two areas (green/paler) on chromosome 15 that are very similar and found in two copies about 3 mb apart. In correct alignment, the two copies of the genes (darker blue) should be aligned.

Second part shows the strand exchange that leads to recombination.

Third part shows the outcome where one chromosome copy has two sets of the genes while the second has zero. It is this second copy that has the area deleted, while the first copy has a duplication. The result of having that duplication is known as "15q11-q13 duplication syndrome (Dup15q syndrome)".

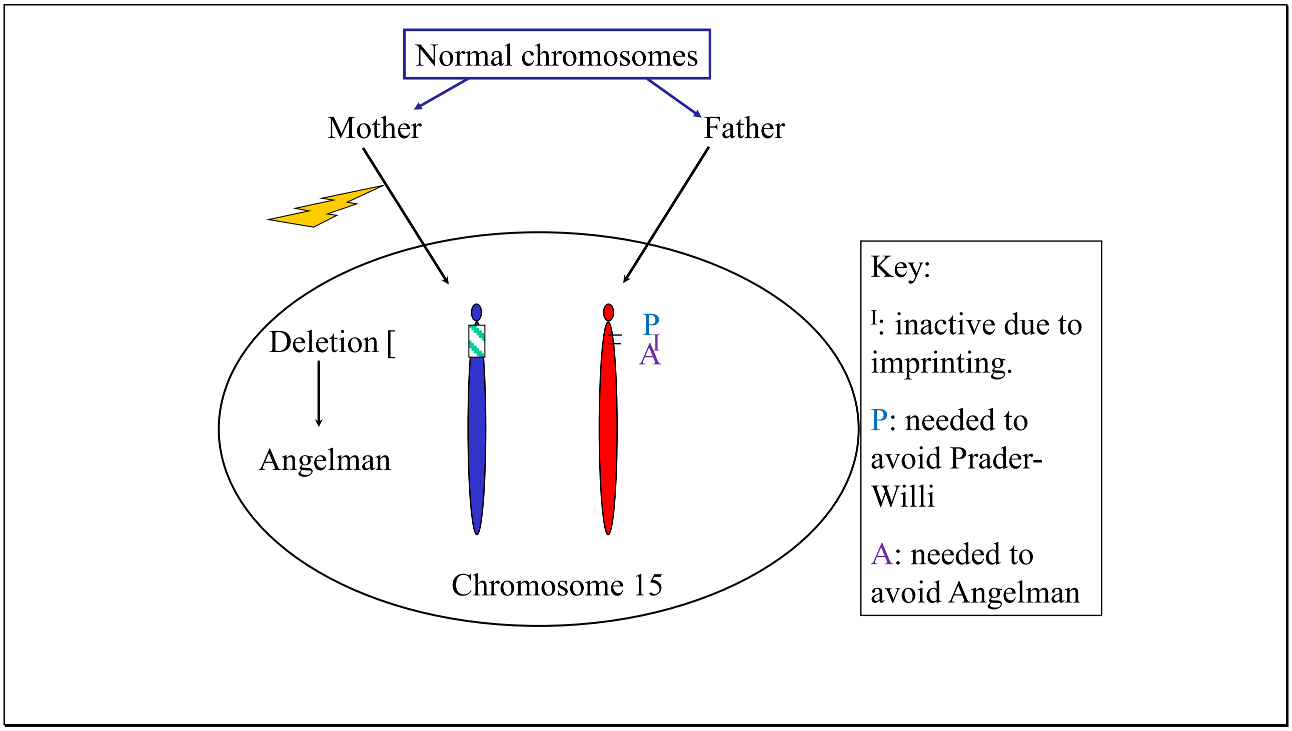
The content of this slide is included because this type of deletion is found in many diseases, and I think it is easier to understand that these deletions occur when seeing the mechanism. Explaining this mechanism is not a learning objective.

If anyone see a good video explaining unequal crossing over at regions of homology, then please forward a link. A link that explains the outcome of unequal crossing over but omits the part of repeated elements being a driver of such crossing over is here: https://youtu.be/xXxuFoz9tWs



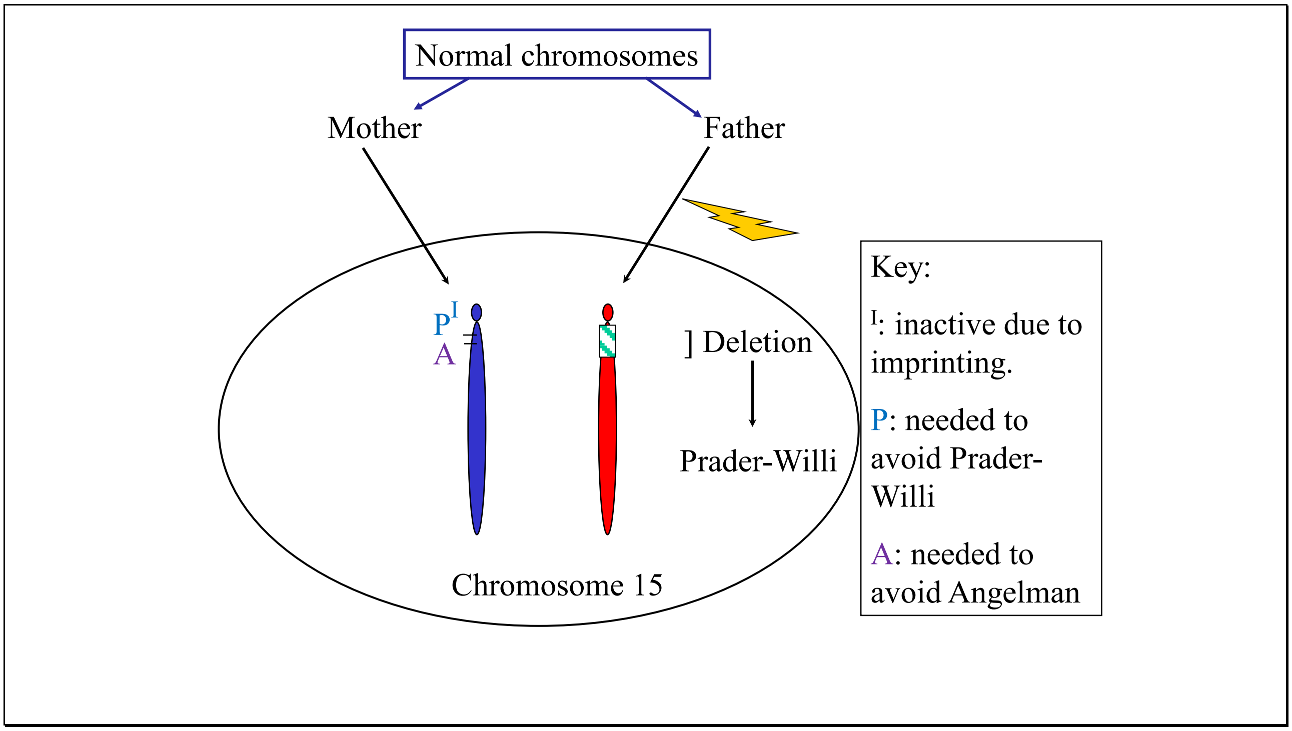
Notice that the letter with superscript I denotes the gene(s) that are imprinted = inactive when coming from that parent. Each person needs one copy of those genes that are not imprinted to protect from the disease.
Notice that it is also possible to stress the opposite side: if the gene for Angelman is imprinted and inactive in the paternal copy, then you can also use the expressetion that it is maternally expressed. Written in that way, then the Prader-Willi genes would be paternally expressed = maternally imprinted and inactive.

Slide 20



For the Angelman gene, only the inactive copy is present, → no protection against that disease

Slide 21



Similarly, no protection against Prader Willi

Slide 22

Disomy

Disomy is very rare, so definitely not familial occurrence

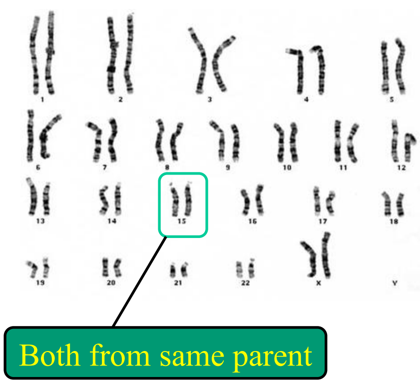
Disomy is a rare event! Prader Willi occurs with an incidence of 1/10000, and only a few percent of these are caused by disomy. However, as a flipside of that: if you have a person with Prader-Willi who does not have a deletion, the next most likely explanation is disomy.

The subtitle's reference to familial occurrence is more relevant for other diseases, not so much for Prader-Willi and Angelman.

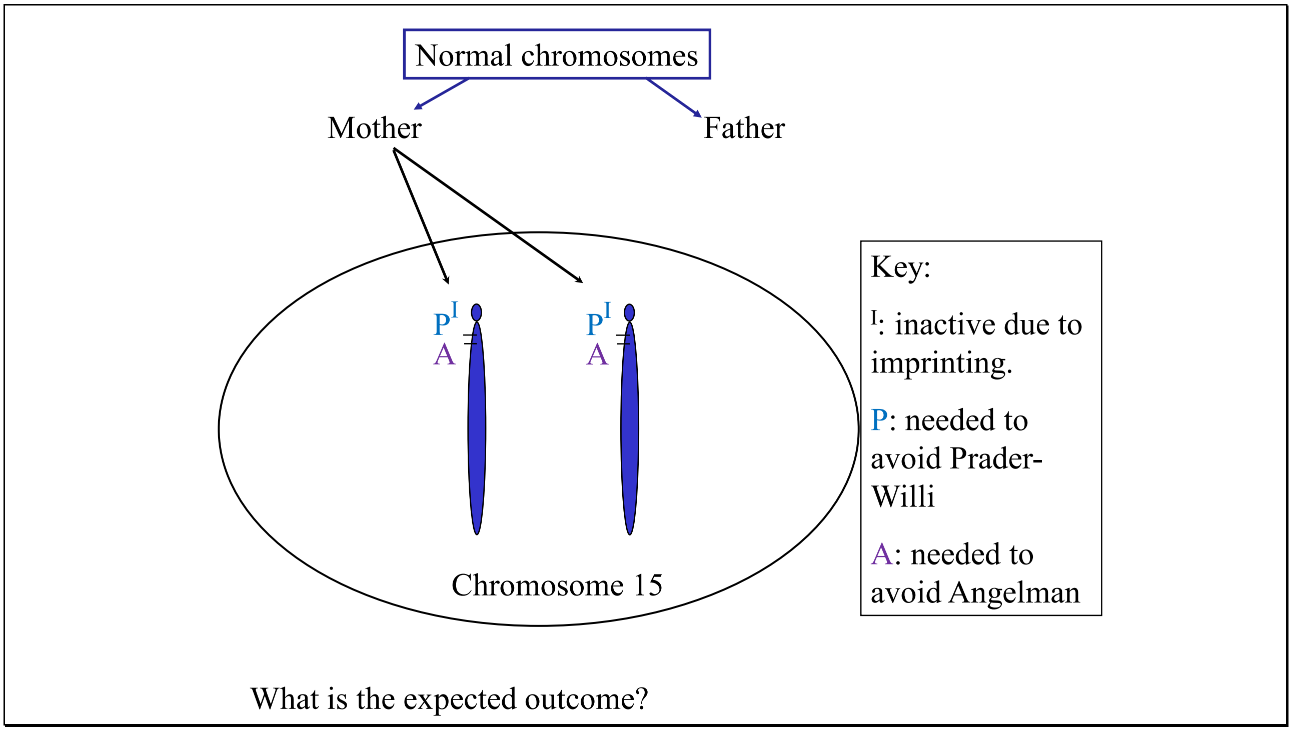
Slide 23

Uniparental Disomy

- Def: Both copies of one chromosome came from the same parent
- (Other chromosomes normal)
- https://www.youtube.com/watch?v=VyJDxYxeMVo minute 15-17 explains disomy



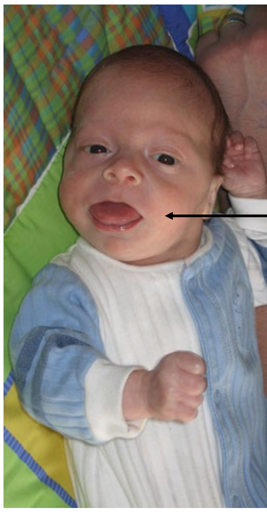
Slide 24



Think about: what is missing? In the case of a maternal disomy, paternal material is missing. In the case of a deletion on a paternal chromosome, paternal material is missing. When Paternal material is missing, the result is Prader Willi. In the cases of maternal material missing (Mat. Deletion/Pat. Disomy) the result is Angelman.

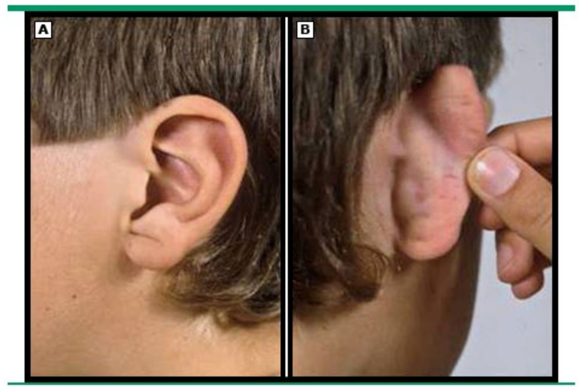
Slide 25

Beckwith-Wiedeman Syndrome



BWS infant with macroglossia

Anterior ear lobe crease and posterior helical ear pits in a patient with Beckwith-Wiedemann syndrome



Images from UpToDate

Slide 26

Beckwith-Weidemann Syndrome

You will hear more about BWS in Repro 2

- This is a syndrome of overgrowth (height >97th percentile)
  - Neonate: omphalocele, macroglossia, visceromegaly, ear pits, adrenocortical cytomegaly, and renal abnormalities
  - Hypoglycemia
  - **Embryonic tumors in childhood** (Wilms tumor, hepatoblastoma, neuroblastoma, rhabdomyosarcoma)
  - Growth may be asymmetrical (Hemihypertrophy)
  - Sometimes normalization of phenotype with age

Hemihypertrophy means asymmetrical (over)growth

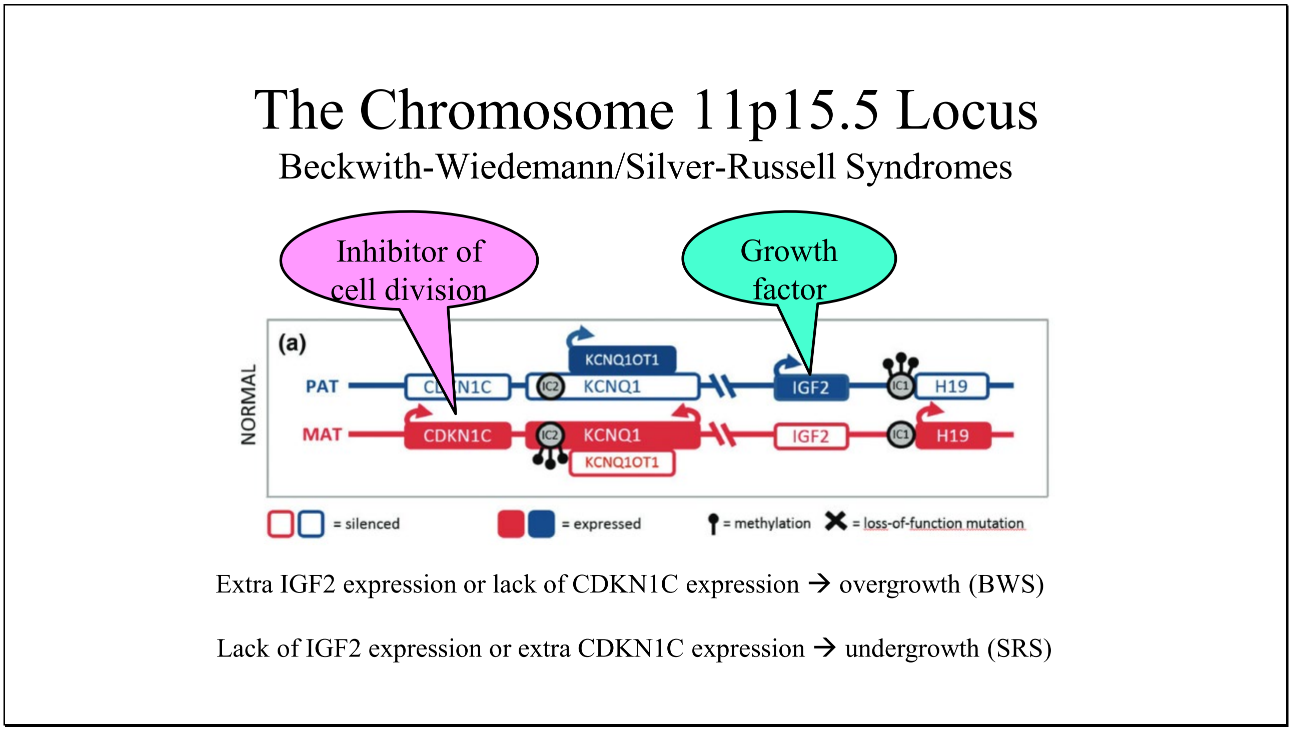
Slide 27

Silver-Russell Syndrome
(aka. Russell-Silver Syndrome)

- Severe pre- and post-natal growth restriction
- Triangular face
  - Prominent forehead
  - Downturn at corner of mouth
- Growth may be asymmetrical
- Mechanism opposite of BWS
  - E.g., disomy would be maternal

.

Slide 28

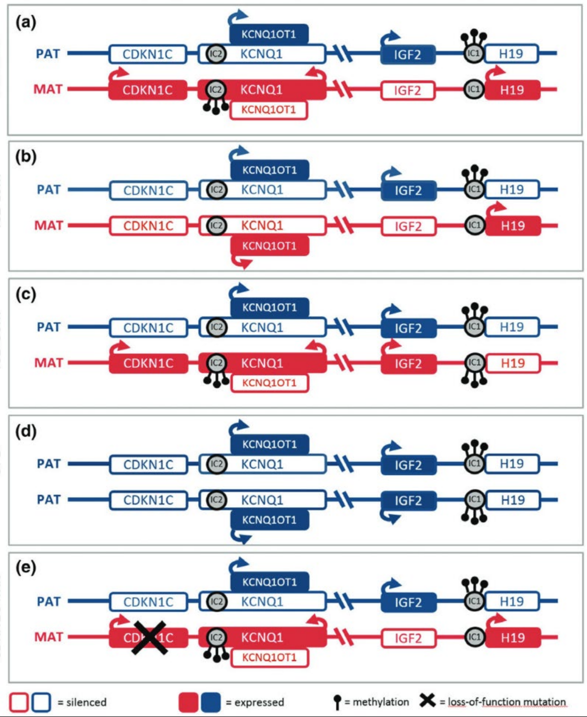


CDKN1C produces the protein p57kip which inhibits cyclin-CDK complexes
IGF2 produces the insulin-like growth factor 2 protein which acts similarly to IGF1 but is more important early in life.
There is some difference in tumor profile depending on which of the two genes are dysregulated.
IC1 and IC2 are two different imprinting centers.
This slide shows the normal situation and is identical to top portion of next slide

Slide 29

Causes of BWS
(b) 50%
(c) 5%
(d) 20%
(e) <1%, can be familial

Abbreviations:
PAT = Paternal
MAT = Maternal
LoM = loss of methylation
GoM = gain of methylation
UPD = uniparental disomy
mut = mutation



https://www.researchgate.net/profile/Alessandro_Mussa2/publication/279728379_Epigenotype-phenotype_correlations_in_Beckwith-Wiedemann_syndrome_a_paradigm_for_genomic_medicine/links/55c4b23208aebc967df37bc1.pdf

The details of how IC2 imprinting influences expression of the different genes is complex and not explained in this lecture. IC1 is explained on later slides.

Slide 30

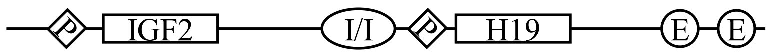
Why Imprinting – One Hypothesis

- A fast growing fetus will come out big and "ready"
  - Fetus w better survival/mother's resources taxed more
  - Paternal genes leading to increased growth is selected for
- Less strong growth of fetus will tax the mother less
  - Mother can live to try another time
  - Maternal genes limiting fetal growth are selected for
- Conflict of interest maternal vs paternal genome
  - This conflict probably stronger in species with higher number of offspring per pregnancy

Paternal ↑ Maternal ↓

Slide 31

The IGF2/H19 area

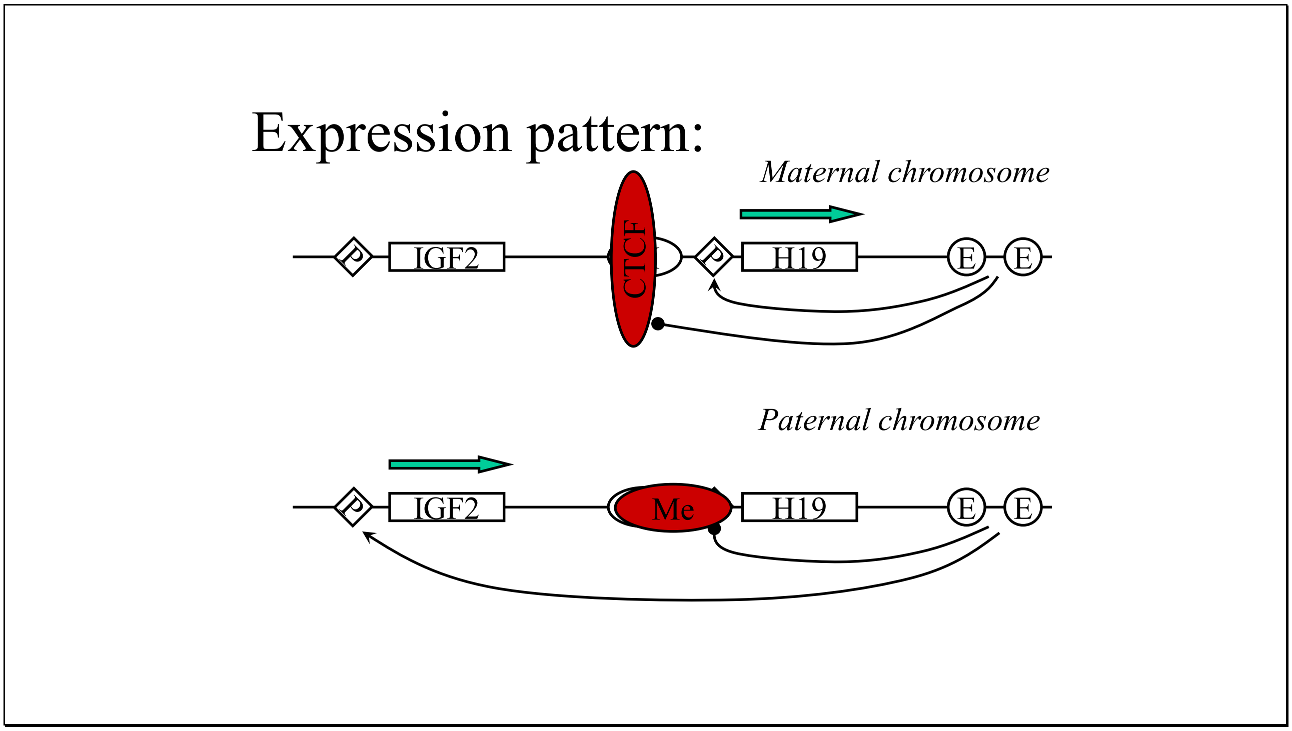


- Legend:
  - IGF2: the area encoding the IGF2 mRNA
  - H19: encodes a non-translated RNA (→ cytosol, complexed with protein)
  - P: promoter region upstream of the transcription unit; not functional without interaction with an enhancer
  - E: enhancer area (several more are known than what is shown here)
  - I/I: Insulator and Imprinting center
  - CTCF: CCCTC-binding factor, a transcription repressor

The I/I included here corresponds to the IC1 shown in previous slides. This is shown differently here to stress that it is an area with dual function.

About the H19: micro-RNA #675 (MIR675) is produced from the H19 transcript by excision. MIRs are regulatory molecules, generally through binding (base-pairing) to mRNAs with complimentary sequence. MIR675 have been shown to influence (probably indirectly) the amount of COL2a1 produced in cultured chondrocytes. The full picture of H19 and MIR675 function is probably still to be discovered, but seems to also involve possible roles in cancer regulation. This note is not expected to result in exam questions.

Slide 32



Maternal chromosome: No methylation → CTCF binds to Insulator (remember, this protein is blocking long distance interactions)

Enhancer

  No interactions w IGF2 promoter → no product

  Interactions w H19 promoter → H19 transcript is produced

Paternal chromosome: Methylation + associated proteins (Me) covers the IC/I → CTCF protein cannot bind

Enhancer:

  No interactions w H19 promoter (not accessible) → no product

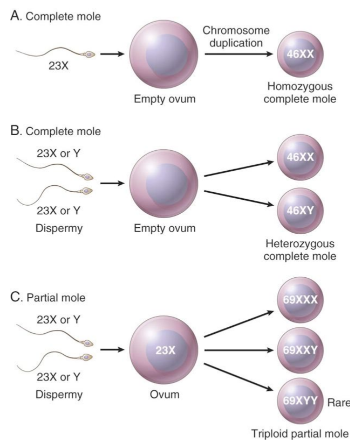
  Interactions w IGF2 promoter → IGF2 transcript is produced

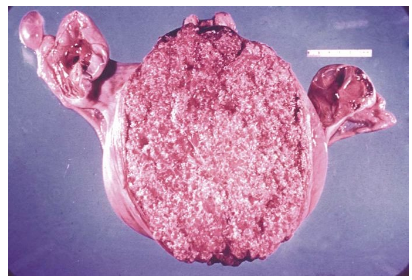
Molar pregnancies show effects of imprinting

Slide 34

Hydatidiform Mole

Remember Sem. 1





Complete hydatidiform mole. Note marked distention of the uterus by vesicular chorionic villi. Adnexa (ovaries and fallopian tubes) are visible on the *left and right side of the uterus.*

Kumar fig 22-53: Kumar, Vinay, Abul Abbas, Jon Aster. *Robbins & Cotran Pathologic Basis of Disease, 9th Edition*. Saunders, 2015. VitalBook file.

From Pathology textbook

Notice the difference between complete and partial moles

Slide 35

Characteristics of Hydatidiform Moles

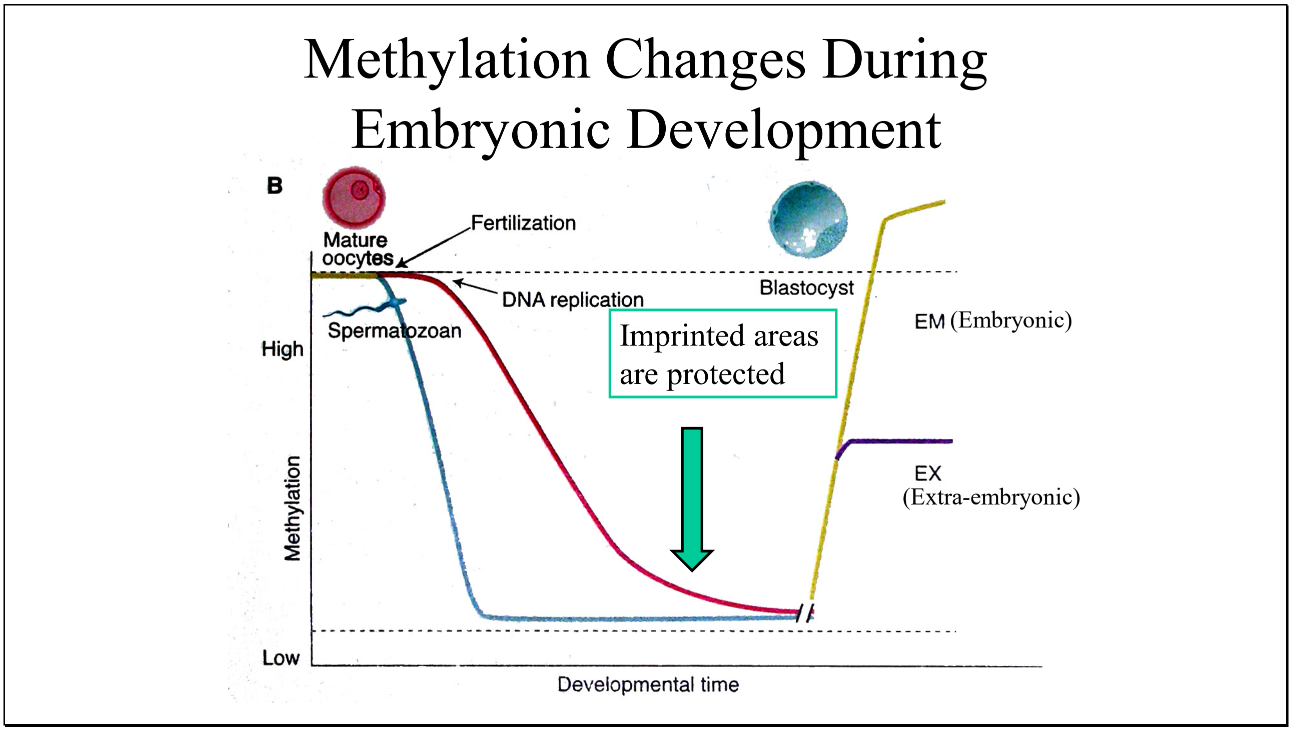
	Complete	Partial
Karyotype	46,XX or 46XY – paternal origin	69,XXY or 69, XYY – 2 sets of paternal chromosomes
Fetal/embryonic tissues	Absent	Present
Hydatidiform swelling of chorion villi	Diffuse	Focal
Trophoblastic hyperplasia	Diffuse	Focal
Scalloping of chorion villi	Absent	Present
Trophoblastic stromal inclusions	Absent	Present
Risk of neoplasia	15-20%	1-5%
Expression of p57	Absent	Present ⟵ Imprinted gene

Data from UpToDate -- Cyclin-Dependent Kinase Inhibitor p57 is **expressed from maternal copy only**

Notice the similarities to some of the other things we have seen

- The complete mole is uniparental, so like a disomy taken to the extreme
- p57 is imprinted and only expressed from the maternal chromosome. It is likely not the only imprinted gene involved but it is the one that is used clinically for diagnosis: if the product is absent then the mole is complete.

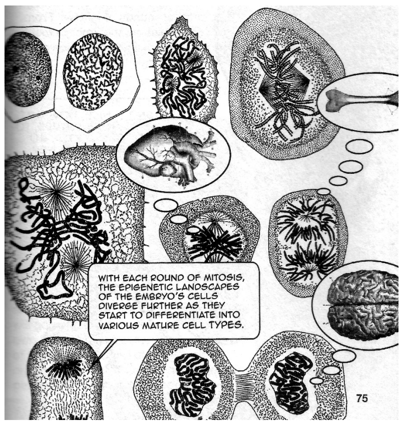
Slide 36



There is a second wave of demethylation after zygote formation, but this demethylation does not affect those areas that are imprinted. Later in embryonic development, new methylation again does not change the imprinting patterns of the chromosomes.
Errors in the processes described here can sometimes lead to disease state in the child (in this slide, mainly if protection of the imprint fails).

Slide 37

FYI: consequences of second round of methylation changes



- Cells start the earliest differentiation
- Differentiated cells differ in where they have methylation and other epigenetic tags

Figure at the left from Ennis and Pugh 2017. Introducing Epigenetics - a graphic guide. Icon books ldt.
This book is an easy read about some of the newer genetics topics.

Epigenetic changes with changed metabolic states

- Recent evidence seems to indicate that metabolic state influence the imprinting of sperm made by a person
  - Obese vs lean men
  - Highly obese vs same man after bariatric surgery
- Also evidence to indicate epigenetic changes to somatic tissues within a person

This is very new. I am not prepared to provide details in this lecture.
There will be no exam questions based on this slide.